



Carbonate microfacies classification model based on dual neural network: a case study on the fourth member of the upper Ediacaran Dengying Formation in the Moxi gas field, Central Sichuan Basin

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Abstract

The dual neural network (DNNW) model combines two neural networks to imitate artificial sedimentary facies division by learning the characteristics of multi-type logging curves corresponding to the sedimentary microfacies of a coring section. The model predicts many non-coring wells in the research zone through logging data. After comparing the classification performance of a fully connected neural network (FCNNW) and AutoML when dealing with three FNDA, EALS, and HRA datasets, the DNNW shows high stability and assists exploration in the Moxi gas field. Four sedimentary microfacies are identified through thin section observation, including thrombolite boundstone (MF1), laminated stromatolite boundstone (MF2), siliceous laminar boundstone (MF3), and micritic dolostone (MF4). Results suggest the fourth member of Dengying Formation in the Moxi gas field is carbonate platform facies deposition: specifically, restricted platform and platform margin.

Keywords Moxi · Fourth member of Dengying Formation · Sedimentary facies · Dual neural network · Well-log

Introduction

The Dengying Formation is the main exploration target of the Sichuan Basin. Presently, more than 20 large- and medium-sized marine gas fields have been found (Liu et al. 2008; Long et al. 2020; Zhao et al. 2020; Guo et al. 2020). Previous studies have found favorable gas accumulation conditions exist in the upper Ediacaran in Moxi (Liu et al. 2008; Long et al. 2020; Zhao et al. 2020; Guo et al. 2020). Until 2019, $5940 \times 10^8 \text{ m}^3$ of gas has been ascertained in fourth member of Dengying Formation (Z_2dn^4 , Member IV), the Moxi gas field (Xu et al. 2018; Jin et al. 2017).

Much progress has been achieved in the exploration of the Moxi gas field, but some geological problems remain to be solved; for example, the petrological discrepancy between the restricted platform and the platform margin, as well as the classification of sedimentary microfacies of the Member IV in this field. Due to the borehole layout of the restricted platform, sedimentary microfacies classification is arduous. To find favorable reservoir microfacies, it is important to make full use of the logging data, core description, and thin section analysis to form a novel template of sedimentary microfacies itemization.

Logging data are a powerful support for sedimentary microfacies partitions, summarizing the relationship between the shapes or data obtained from logging curves and the depositional microphases (Murphy et al. 2000; Liu et al. 2006; Wang, 1991). However, artificial microfacies classification gives rise to inaccuracies of classification, restricting the reliability of oil and gas exploration.

With the advent of artificial intelligence, algorithms have become a new method for sedimentary microfacies division. Much of the research in sedimentology in the last two decades has examined neural network (NNW) model preferences, using backpropagation (BP) neural networks to learn the logging data and identify sedimentary microfacies (Xu and Chen 2001;

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Ma 2017), fuzzy inference neural networks to simulate the thinking process of logging engineers identifying sedimentary microfacies (Liu and Xu 2017; Wang and Mei 2008; Jin et al. 2006), and image model networks to identify sedimentary microfacies (Ren et al. 2019; Sun et al. 2009). Past successful research indicates algorithms are suitable for dividing clastic rocks' sedimentary microfacies and sometimes also work for dividing carbonate rocks' sedimentary microfacies utilizing only one curve (GR) or two curves (GR and RT) (Satyanarayana et al. 2007; Ojha and Maiti, 2016; Valentín et al. 2019; Zheng et al. 2021). However, traditional methods have a poor performance to classify microfacies in Moxi gas field. To solve this problem, this study compares FCNN predicting results and testified model structures with different datasets (Fig. 1).

The purpose of this study is to classify sedimentary microfacies in the Moxi gas field with an improved neural network modeling algorithm, the dual neural network (DNNW), and reconstruct the sedimentary microfacies.

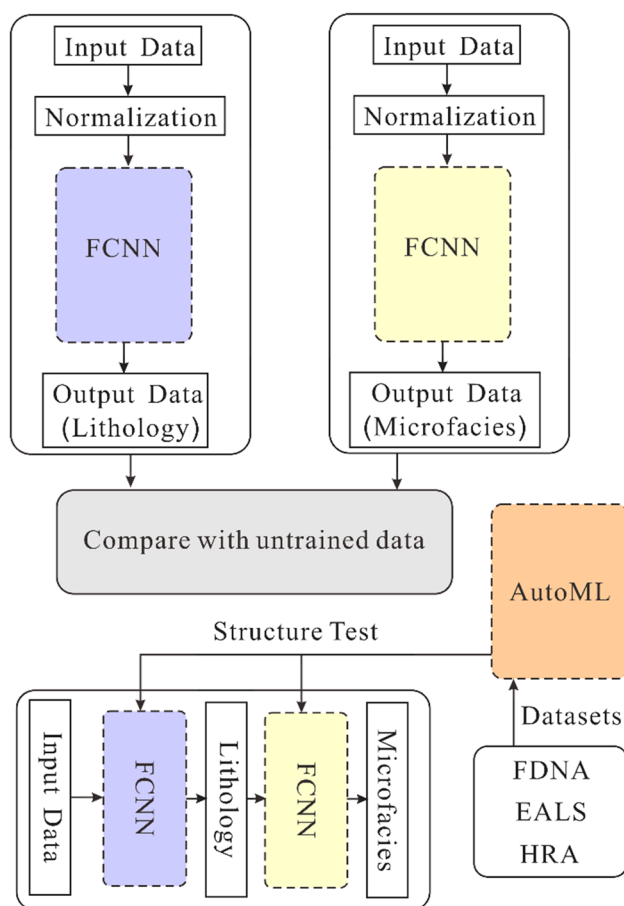


Fig. 1 Workflow of test and generating of DNNW

Geological settings

The Sichuan Basin is located in the western Yangtze block (Fig. 2a), and the research zone is located in the central Sichuan Basin (Fig. 2a).

During the late Ediacaran, Yangtze block was in an extensional tectonic regime (related to Rodinia supercontinent broke up), which is called Xingkai taphrogenesis (Craig et al. 2009; Lottaroli et al. 2009; Huang et al. 1980; Li et al. 2008; Liu et al. 2013; Luo 1981, 1984; Sun et al. 2011). Associating with subsequent Tongwan movement, the Dengying Formation (Z_2dn) overlies the Sinian Doushantuo Formation (Z_2d) and the Cambrian Qiongzhusi Formation (ϵ_1q), showing unconformity contact with the Qiongzhusi Formation (Fig. 2c) (Hou et al. 1999; Liu et al. 2015; Wang et al. 2014). After serious tectonic uplifts, central Sichuan Basin was in stable stage and a platform with dual platform margin developed in the Sichuan Basin (Chen et al. 2017). The research zone situated in the western platform margin (Chen et al. 2017). In this environment, jillion microbial bred in the relatively high physiognomy and developed microbial mound-shoal complex, which is an essential prospecting target (Qian 1991; James and Dalrymple 2010; Guhey et al. 2011).

The Dengying Formation was found in the Dengying strait, Three-Gorge region (Lee and Chao 1924). The preceding Dengying Formation included the Liantuo Formation, Nantuo Formation, Doushantuo Formation, and Dengying Formation, while the new Sinian Dengying Formation was redefined by the National Commission on Stratigraphy, corresponding to Ediacaran (635–542 Ma) (Xing and Sun 1989; Condon et al. 2005).

Previous studies indicate that the Dengying Formation can be divided into four members, Member I (Z_2dn^1), Member II (Z_2dn^2), Member III (Z_2dn^3), and Member IV (Z_2dn^4), by their content of cyanobacteria (Zou et al. 2014; Liu et al. 2016; Song et al. 2018). Large amounts of cyanobacteria are found in Member II and Member IV, forming laminated, stromatolite, and thrombolite fabrics (Zou et al. 2014; Liu et al. 2016). Member I and Member III lack cyanobacteria while Z_2dn^2 and Z_2dn^4 enrich cyanobacteria (Zou et al. 2014; Liu et al. 2016; Song et al. 2018).

The Member IV in the Moxi gas field is composed of microbial laminar dolostone and microbial laminated dolomite, with a large number of thrombolite dolomites (Wei et al. 2015; He et al. 2020; Li et al. 2019; He 2013; Liu et al. 2009). In some areas, micritic dolomite developed, implying a restricted platform sedimentary environment which includes the microbial flat, granular shoal, and platform flat (Fig. 2c). Accompanying karstification in the Member IV, the microbial mound-shoal subfacies is well developed in the

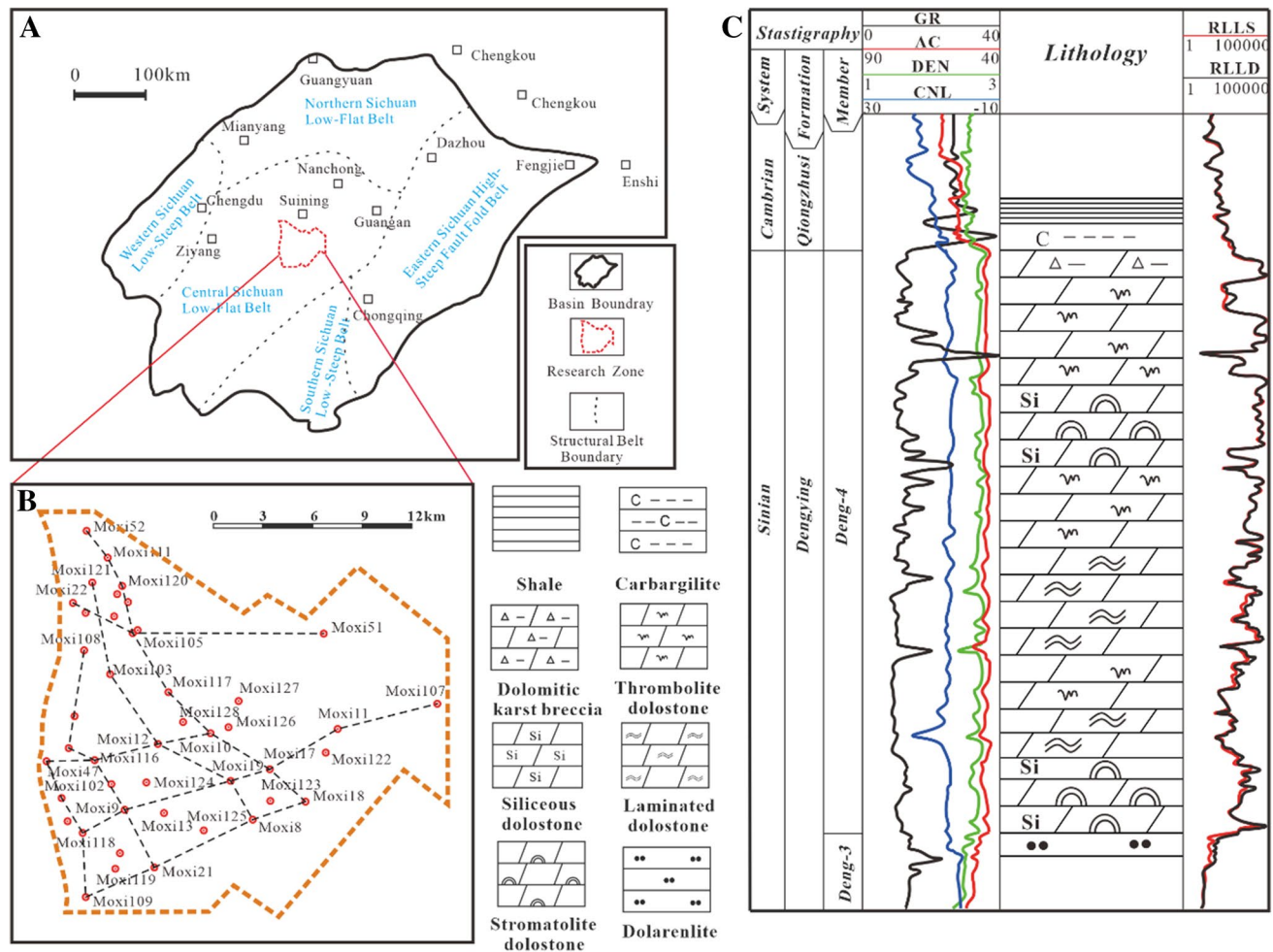


Fig. 2 Location and comprehensive lithologic columns in research zone. **A** The location of Sichuan Basin and research zone. **B** Location of wells and connecting-well profile orientation in Moxi. **C** Comprehensive lithologic columns of Member IV

research zone, composed of laminated dolostone, stromatolite dolostone, and thrombolite dolostone (Wei et al. 2015; He et al. 2020; Li et al. 2019; He 2013; Liu et al. 2009). Considering sedimentary cycle and well logging data, Member IV can be divided into six sublayers, including 1–1 layer, 1–2 layer, 1–3 layer, 2–1 layer, 2–2 layer, and 2–3 layer.

Methods

Core and thin section sample

Many wells are drilled in the Moxi gas field, containing Moxi9, Moxi21, and Moxi108, etc. (Fig. 1b). According to the core data, the main depth of Member IV in the Moxi gas field is between 5000 and 5450 m, and the petrology is dominated by dolomite, laminated dolostone, stromatolite dolostone, and thrombolite dolostones. Sand debris, calcareous

and siliceous materials are also included, with the main microfacies being thrombolite and laminated boundstone.

Logging data

After performing the correlation, as in the test chart (Fig. 3g), eight kinds of logging data were used in this study, including the acoustic velocity (AC), borehole diameter (Ribeiro et al.), neutron (CNL), density (DEN), gamma-ray (GR), photoelectric absorption cross-section index (PE), deep lateral resistivity logs (RLLD), and shallow resistivity logs (RLLS).

Dual neural network

Normally, an NNW model has a unique type of input data and sole type of output layer architecture (Xu and Chen 2001). Past instructive sedimentary microfacies identification models used logging data as the input and

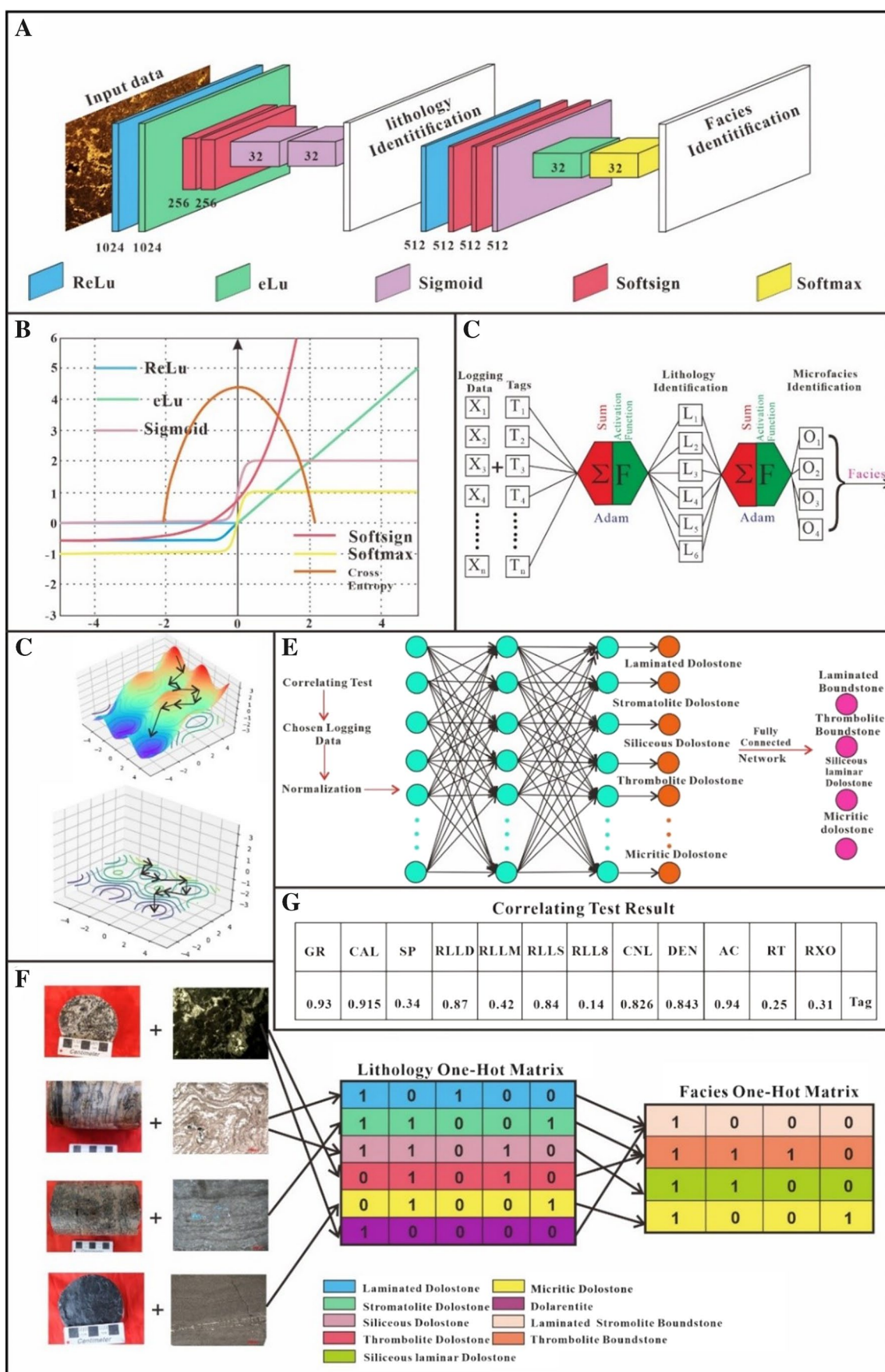


Fig. 3 All methods used in the neural network algorithms. **A** The structure of DNNW modeling (Activation functions and neural unit's numbers are shown). **B** All activation and loss functions' graphs. **C** Illustration of artificial network. **D** 3-dimensional diagrammatizing of Adam optimizer. **E** Structure of fully connected network every neural unit will connect with all neural units in the next layer. **F–G** Tags and two one-hot coding matrices (lithology and facies matrix)

sedimentary microfacies data directly as output (Xu and Chen 2001; Ma 2017; Liu and Xu 2017; Wang and Mei 2008; Jin et al. 2006; Zheng et al. 2021). However, sedimentary microfacies refer to the smallest units, each with a unique rock type, structure and rhythm (Wang 1979; Zeng et al. 1979; Liu 1987). Accordingly, ignoring lithological data and using microfacies directly as output data violates the definition of microfacies. Therefore, a new method (DNNW) is put forward to replace the previous fully connected (FCNNW) model. The DNNW is designed to imitate the process of sedimentary facies classification artificially. With two FCNNW models (Fig. 3a, c, e), lithology data in each well are identified first, and lithological identification takes the role of input data, flowing into the second FCNNW to predict sedimentary microfacies (Fig. 3a, c, e).

Learning samples

Learning samples are several sorts of data where the specific calculating relationship is unclear; i.e., logging data are reflections of lithology and microfacies while a specific formula is hard to reach. In this study, learning samples were made with a core and thin section observation complex and logging data. From the thin section and core observing data, lithology and sedimentary facies can be identified and used in the role of tags.

Shuffle algorithms

During the learning process, the neural network would be captured in the local optimal solution without reshuffling the sample. Despite the high degree of accuracy, captured data actually disrupt the construction of reflections between other samples. Here, the shuffling algorithm is implemented in three steps:

- (1) Randomly change the order of all the examples using Python;
- (2) Calculate types of learning samples and compute mathematical expectations and variance
- (3) Extract the learning samples by numerical features, as follows:

$$E\left(\sum_{k=1}^n a_i + c\right) = \sum_{k=1}^n a_i E(x_i) + c \quad (1)$$

$$D\left(\sum_{k=1}^n a_i\right) = E\left[\left(\sum_{k=1}^n a_i\right)^2\right] + \left[E\left(\sum_{k=1}^n a_i\right)\right]^2 \quad (2)$$

Activation function

The activation function is an essential tool to realize non-linearization among those neurons (Zhang, 2006). This study uses four activation functions, including ReLU (rectified linear unit), ELU (exponential linear units), Sigmoid, and Softmax (Fig. 3b).

The ReLU function is a normal slope function-achieving non-linear function to magnify the variance of bias and filter out negative bias (Fig. 3b) (Xu et al. 2015).

$$f(x) = \begin{cases} 0, & x < 0 \\ x, & x \geq 0 \end{cases} \quad (3)$$

The ELU function is an improved activation function that, contrasting the ReLU function, keeps outputting with correlating negative input (Fig. 3b). Overwhelming the ReLU function, the ELU function usually has stronger robustness (Daeho et al. 2020; Grelsson and Felsberg 2018; Oquab et al. 2014; Dahl et al. 2013; Mandal and Sarpeshkar 2007).

$$f(x) = \begin{cases} x, & x > 0 \\ a(e^x - 1), & x \leq 0 \end{cases} \quad (4)$$

The Sigmoid function is a normal function used in neural network algorithms (Fig. 2b); its formulation is (Salakhutdinov and Hinton 2009):

$$f(x) = \frac{1}{1 + e^{-x}} \quad (5)$$

The Softmax and Sigmoid functions converge and trim the output units (Fig. 2b) (Salakhutdinov and Hinton 2009; Seila 2007):

$$Softmax(x_i) = \frac{e^{x_i}}{\sum_{k=1}^i e^{x_k}} \quad (6)$$

Loss function

The loss function evaluates the difference between real data and predicting data. Here, this study adopts a cross-entropy error function (cross-entropy) as the loss function (Fig. 3b). Cross-entropy describes the distance between the

actual output and the expected output. The smaller value of cross entropy means a closer relationship between several distributions (Kingma and Ba 2014; Shore and Johnson 1980).

$$L = \frac{1}{N} \sum_i L_i = -\frac{1}{N} \sum_i [y_i \cdot \lg(p_i) + (1 - y_i) \cdot \lg(1 - p_i)] \quad (7)$$

Optimizer

An optimizer is a mathematical tool that adjusts the weight and bias of neural units to reach an appropriate status, normally high accuracy and low loss, by training.

This study applies the optimizing algorithm, Adam (Kingma and Ba 2014) (Fig. 3d). The advantages of using this method are (1) high computational efficiency; (2) minimal memory requirements; (3) that the diagonal rescale of the gradient is invariant (Adam multiplies the gradient by a diagonal matrix with only positive factors, using the stack swapping method); (4) suitability for problems with data or parameters; (5) suitability for fixed targets; (6) suitability for very noisy and/or sparse gradient problems; and (7) hyperparameters have intuitive interpretation and usually require little adjustment.

$$\begin{aligned} m_t &= \beta_1 m_{t-1} + (1 - \beta_1) g_t \\ v_t &= \beta_2 v_{t-1} + (1 - \beta_2) g_t^2 \end{aligned} \quad (8)$$

$$\hat{m}_t = \frac{m_t}{1 - \beta_1^t}, \hat{v}_t = \frac{v_t}{1 - \beta_2^t} \quad (9)$$

$$W_{t+1} = W_t - \frac{\eta}{\sqrt{\hat{v}_t} + \epsilon} \hat{m}_t \quad (10)$$

Net structure

To fully imitate the process of sedimentary facies division by the human brain, this study established a dual neural networking structure (Fig. 3a, c, e). Toward this goal, two tags (lithology and sedimentary microfacies) are attached to the learning samples. At the same time, the network is made up of two joint functional modules. After comparing and examining the model (see the “Modeling results” section), the first neural network predicts lithology by training, while the second utilizes a similar structure to construct relationships between lithological results and sedimentary facies. The connecting path between the two modules is the logging data. Logging data take on roles as input data. After the first

training, during the second network training, the input data are the lithological results (predicted by the first network), while the logging data become part of tags to supervise the training process with microfacies tags (Fig. 3a).

Results

Geological results

Lithological and microfacies

Combining limited thin sections and core data, the Member IV in the Moxi gas field can be described as below.

From core-drilling data from Moxi8, Moxi9, Moxi13, Moxi23, and Moxi105, the lithology of the lower part of the Member IV is stromatolite dolostone, siliceous, and a little dolarentite. At the bottom, the stromatolite dolostone (Fig. 4h) and laminated dolostone are usually silicified. The thickness of the lower segment of Member IV in the Moxi gas field is generally 8–30 m, with an average thickness of about 32.38 m (Table 1).

The lithology of the upper part of the Member IV is mostly composed of dolomite karst breccia (Fig. 4o), thrombolite dolostone (Fig. 4a, b, c, h), stromatolite dolostone (Fig. 4h), and micritic dolostone (Fig. 4j, k, l). The structure of the microbial mound is developed (Fig. 4m). The thickness of the upper part of Member IV in the Moxi gas field is generally between 100 and 250 m, with an average thickness of 224.34 m (Table 1).

Composed of laminated dolostone and stromatolite dolostone, the laminated stromatolite boundstone microfacies are shown as a bright lamination and a dark layer (Fig. 4g, h); this usually develops in the supratidal zone. Laminated stromatolite boundstone microfacies are millimeter-grade, and the internal composition of the lamination is different. The fine-grained layer is a micrite layer, and the coarse-grained layer is mudstone calcite. Horizontally, the microbial lamination can be continuous or intermittent, and the interlayer is usually a light-colored thin layer of micritic dolomite (Fig. 4g, h) which is formed by mucus released by microorganisms adsorbing sediments (Jin et al. 2017).

Thrombolite boundstone microfacies are composed of light gray mud dolomite and grains (Fig. 4a, b, c). The thrombolite dolostone does not have a laminar fabric, often condensed by cyanobacteria and green algae to form a thrombolite structure. Dark algae-rich deposits are spongy and cloudy, being symbiotic with micritic dolostone and stromatolite dolostone. On a microscopic scale, the thrombolite structure has unique growth orientations of microorganisms, as the characteristics of cyanobacteria are controlled by the growth mode, mineralization, trapping, adhesion of cyanobacteria, and the surrounding environment (Jin

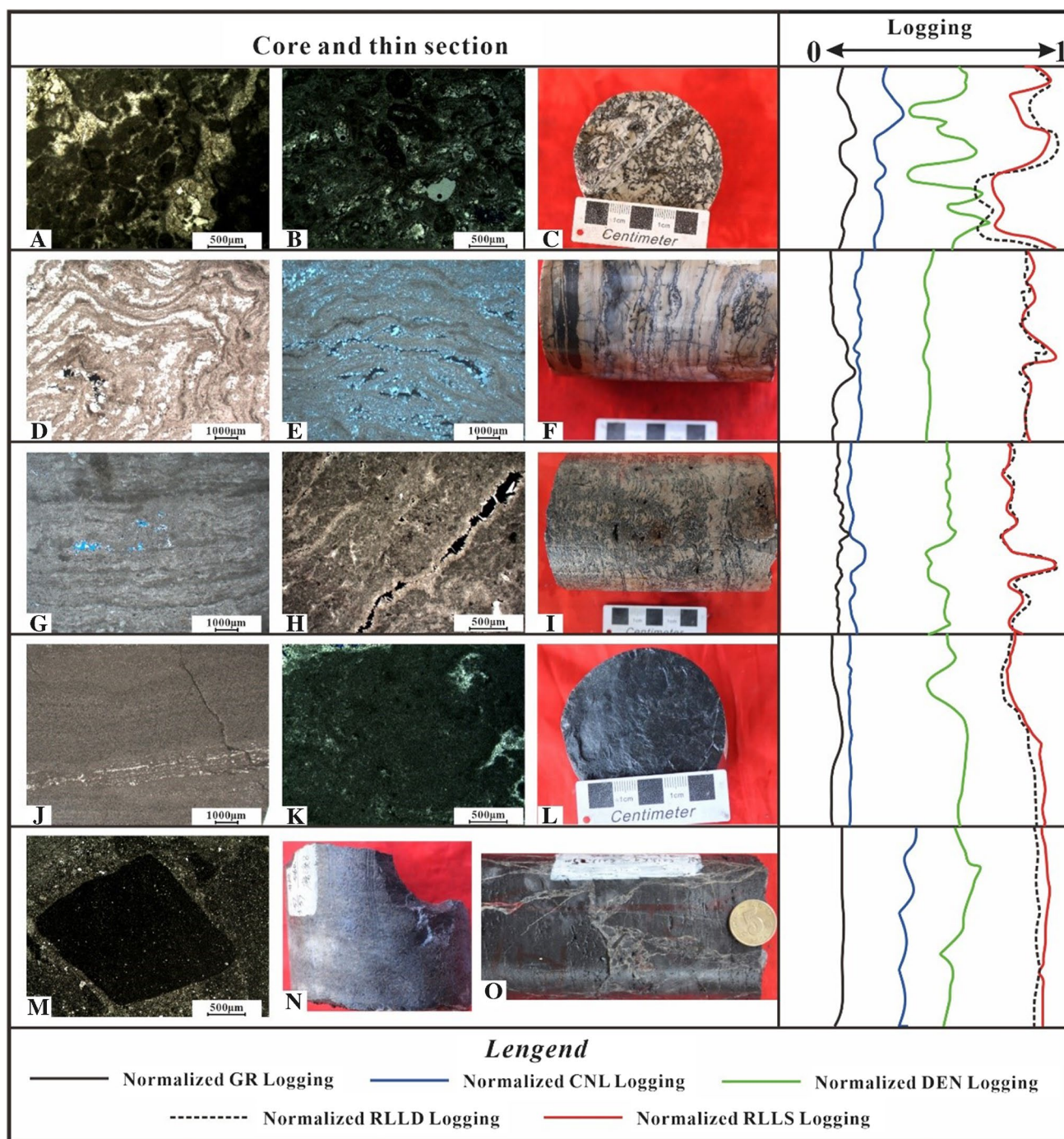


Fig. 4 Results of core observation, microscopic observation, and normalized logging data. **A.** Thrombolite dolostone, MX105-3, 5303.7 m, 5×, PPL. **B.** Thrombolite dolostone, MX105-3, 5303.7 m, 5×, XPL. **C.** Thrombolite dolostone, MX23, 5210.8 m. **D.** Siliceous dolostone, MX51, 5334.76 m, 5×, PPL. **E.** Siliceous dolostone, MX105, 5360.58 m, 5×, XPL. **F.** Siliceous dolostone, MX8, 5114.76 m. **G.** Laminated dolostone, MX105, 5346.48 m, 5×, PPL. **H.** Laminated dolostone, stromatolite dolostone, MX102, 5-19/50,

5346.48 m, 5×, XPL. **I.** Light gray stromatolite dolostone, laminated dolostone, MX105, 5345.82–5346.98 m. **J.** Micritic dolostone, MX51, 5369.56 m, 5×, PPL. **K.** Micritic dolostone, MX105-8, 5369.56 m, 5×, XPL. **L.** Micritic dolostone, MX13, 55,050.34 m, 5×. **M.** Breccia dolostone, MX39-1-1B, 5310.98 m, 5×, PPL. **N.** Medium grey sand gravel dolostone, MX8, 5160.2 m. **O.** Medium grey breccia dolostone, MX23, 5214.91 m

Table 1 Statistics of layer thickness of member IV

	Top (m)	Bottom (m)	All (m)	Upper (m)	Lower (m)
Moxi8	5087	5342.26	255.26	34.2	221.06
Moxi9	5002.4	5324.295	321.895	51.4	270.495
Moxi13	5015.59	5290.323	274.733	22.7	252.033
Moxi23	5208.2	5323.854	115.654	8.63	107.024
Moxi105	5294	5610	316	45	271
Average			256.7084	32.38	224.34

et al. 2017). In the Moxi gas field, thrombolite boundstone microfacies are rich in organic matter, and most are well preserved, reflecting the medium and high energy.

The siliceous laminar boundstone microfacies are characterized by thin- to medium-thickness layers, with a single layer thickness of 2–10 cm, striped and banded structure. On a microscopic scale, the laminar fabric of siliceous laminar boundstone microfacies is characterized by cryptocrystalline silicate, crystalline granular silicate, radial chalcedony, and petal siliceous nodules (Fig. 4d, e, f). The microfacies are formed by the metasomatism of primary laminated stromatolite boundstone by hydrothermal fluid.

Micritic dolostone microfacies (Fig. 4j, k, l) are fine to very-fine micritic dolomite on a microscopic scale.

Sedimentary facies

Based on the core observation, thin section analysis, and paleogeographic research (Chen et al. 2017; Chen et al. 2019), the sedimentary facies system division is as follows:

The platform margin (corresponding to standard facies FZ5 by Flügel 2010) is composed of microbial mound-shoal complex subfacies and has more thrombolite dolostone, laminated dolostone, and stromatolite dolostone than restricted platform. Microbial mound-shoal complex subfacies is made

up with thrombolite boundstone (MF1) and laminated stromatolite boundstone (MF2).

The restrict platform (corresponding to standard facies FZ7 by Flügel 2010) is composed of microbial shoal and interbank sea subfacies, showing less microbial fabric than platform margin. Microbial shoal subfacies include siliceous laminar boundstone (MF3). Interbank sea subfacies include micritic dolostone (MF4).

MF1 equals to SMF7 (Flügel 2010), containing thrombolite dolostone. MF2 equals to SMF20 (Flügel 2010), mostly containing laminated dolostone and stromatolite dolostone and partially dolarenite. MF3 is a special instance of MF2. After influxes of hydrothermal fluid, MF2 switches into MF3, exhibiting more siliceous dolostone. MF4 equals to SMF23 (Flügel 2010), containing a huge amount of micritic dolostone (Table 2).

Modeling results

Comparison

To compare the DNNW and FCNNW model performances, this study used several datasets from Kaggle (<https://www.kaggle.com/datasets>), including Fishing Data North Atlantic (FDNA) (<https://www.kaggle.com/alexanderbader/fishing-data-north-atlantic>), End ALS Kaggle Challenge (EALS) (<https://www.kaggle.com/alsgroup/end-als>), and HR Analytics: Job Change of Data Scientists (HRA) (<https://www.kaggle.com/arashnic/hr-analytics-job-change-of-data-scientists>). Furthermore, we designed the AutoML method to measure the rationality of both the DNNW and NNW. After measurement, parameters of net structure are set. During the test, the DNNW opts the result of the middle layer as input data to ensure similarity with the process in microfacies' identification. AutoML has a similar structure to DNNW and FCNNW, and concrete parameters of the structure would

Table 2 Sedimentary facies division of member IV

Facies	Standard facies by Flügel 2010	Subfacies	Microfacies	Standard microfacies by Flügel 2010	Lithology
Platform margin	FZ5	Microbial mound-shoal complex	Thrombolite boundstone (MF1)	SMF7	Thrombolite dolostone
			Laminated stromatolite boundstone (MF2)	SMF20	Laminated dolostone
					Stromatolite dolostone
Restrict platform	FZ7	Microbial shoal	Siliceous laminar boundstone (MF3)	SMF20	Dolarenite
					Siliceous dolostone
					Laminated dolostone
					Stromatolite dolostone
		Interbank sea	Micritic dolostone (MF4)	SMF23	Dolarenite
					Micritic dolostone

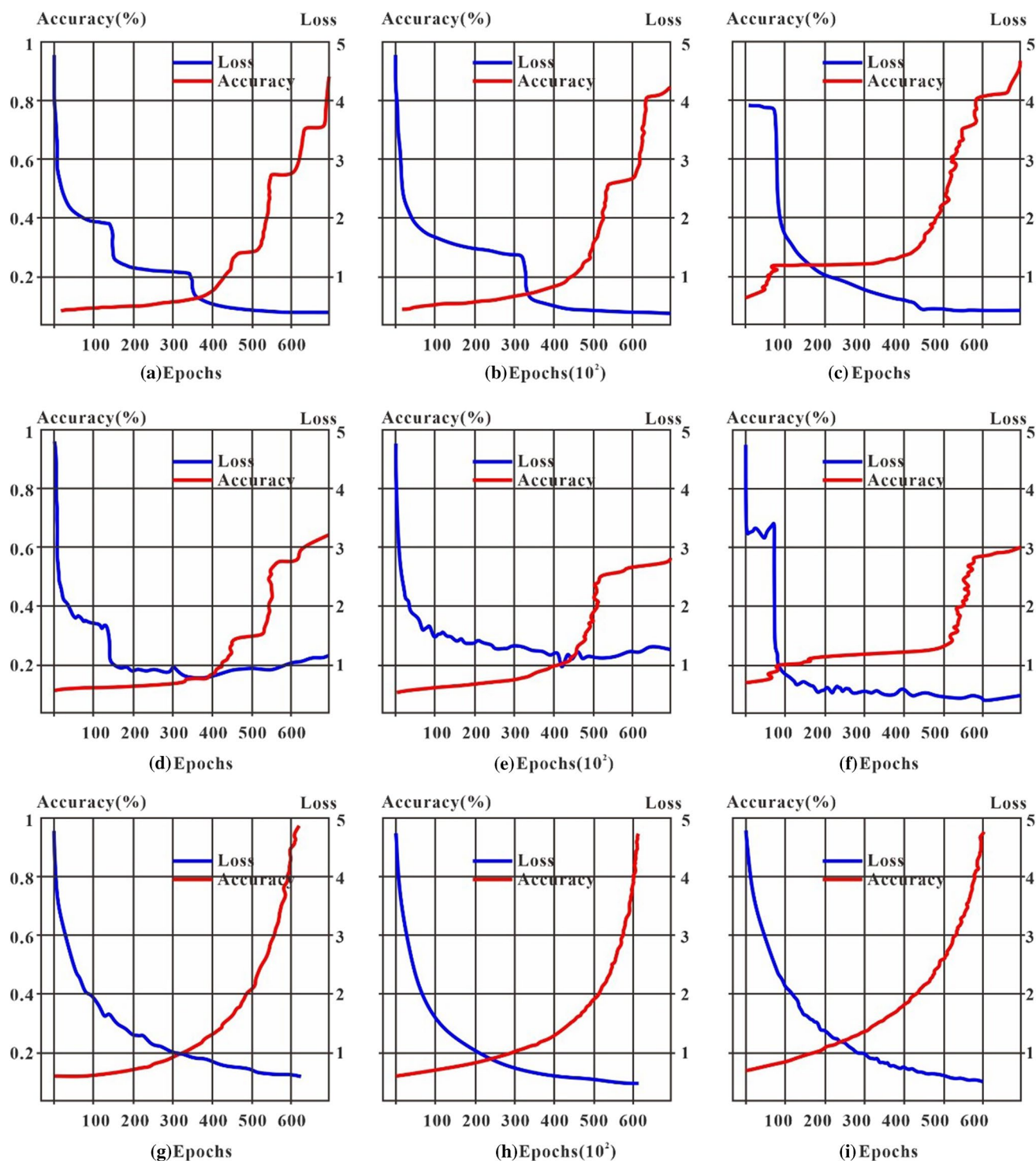


Fig. 5 Diagrams of net architecture test. **A** Classification test of FNDA by DNNW. **B** Classification test of EALS by DNNW. **C** Classification test of HRA by DNNW. **D** Classification test of FNDA by NNW. **E** Classification test of EALS by NNW. **F** Classification test of

HRA by NNW. **G** Classification test of FNDA by AutoML. **H** Classification test of EALS by AutoML. **I** Classification test of HRA by AutoML

be trained automatically by AutoML. Looking at the test performance, DNNW and AutoML have higher accuracy for three datasets (Fig. 5), and AutoML shows a similar loss and accuracy when classifying these three datasets (Fig. 5).

Logging data pre-processing

The statistical results of tests of logging data integrity show that only some logging curves in the area have high integrity which is suitable for logging prediction (Table 3).

In the process of the logging operation, some logging data were missing or invalid due to environmental interference, improper manual operation and logging tool failure. Tensorflow/pandas processing shows some missing and invalid values filling in logging data of Member IV in Moxi gas field. The missing well section includes 5016–5036 m PE, RT, and RXO logging curves of the Moxi13 well; invalid value filling includes the RT curve of the Moxi13 well (depth: 5042.375–5045.375 m) and RXO logging curve of the Moxi17 well (depth: 5073.625–5079.25 m) (Table 4).

Normalization

The normalization calculation method is used to eliminate numeric differences between logging data, i.e., RT curves often vary between 10,000 and 20,000, while GR curves vary between 0.1 and 0.5, which brings huge errors to network training. Here, the formula is as follows:

$$P_i = \frac{L_i - L_{m,n}}{L_{max} - L_{m,n}} \quad (11)$$

After normalization, data from different logging curves are mapped to [0,1] topological space. Among them, P_i is the normalized logging data value, L_i is the logging data value before normalization, $L_{m,n}$ is the minimum value of the statistical results of the same attribute logging data, and L_{max} shows the maximum statistical result of the same attribute logging data. The normalized processing results of logging data in the Moxi gas field are shown in Table 2.

One-hot coding

The sedimentary microfacies in the research zone are closely related to the lithology of the core section, while the sedimentary microfacies determined by lithology are text data. The program in the neural network model can only process digital data, so data conversion of sedimentary microfacies is particularly critical. One-hot coding is an effective method to transform data class attributes in classification problems. The principle of independent one-hot coding is to take the number of data classes as the dimension number of topological space. In a certain data dimension, 0 means refusing to give the unit step size of data topological space under the dimension, and 1 means agreeing to grant the unit step size of data topological

Table 3 Statistics of logging curve integrity of Member IV

Curves	AC	CAL	CNL	DEN	GR	PE	RT	RXO
Integrity	100%	100%	100%	100%	100%	94%	91%	95%

Table 4 Normalized results of some logging curves in the Member IV

Curves	AC	CAL	CNL		PE	RT	RXO
Normalization	0.92	0.069	0.22		0.054	0.016	0.098
	0.86	0.145	0.15		0.52	0.85	0.24
	0.21	0.210	0.11		0.62	0.42	0.098
	0.52	0.089	0.09		0.29	0.34	0.56
							

Table 5 Result of one-hot coding of lithology in Member IV

Lithology	One-hot Code						
Laminated dolostone	1	0	1	0	0	0	0
Stromatolite dolostone	1	1	0	0	1	0	0
Siliceous dolostone	1	1	0	1	0	0	0
Thrombolite dolostone	0	1	0	1	0	1	1
Micritic dolostone	0	1	0	0	1	1	1
Dolarenite	1	0	0	0	0	0	1

Table 6 Result of one-hot coding of facies in Member IV

Sedimentary microfacies	One-hot code			
Laminated stromatolite boundstone	1	0	0	0
Thrombolite boundstone	1	1	1	0
Siliceous laminar boundstone	1	1	0	0
Micritic dolostone	1	0	0	1

space under the dimension. The computer can code certain sedimentary microfacies by calculating the Euclidean space distance of the data. The results of one-hot coding of sedimentary microfacies and lithology in coring sections are shown in Tables 5 and 6.

Training

After 5000 times training for lithology, the cumulative loss of the training set is 1.7532, and the training accuracy is 84.23%. The cumulative loss of the validation set is 1.6068, and the accuracy of validation is 83.23%. Subsequently, for microfacies, with 5000 training epochs, the cumulative loss of the training set is 1.8421, and the training accuracy is 88.45%. The cumulative loss of the validation set is 1.7323, and the accuracy of validation is 83.23% (Fig. 6a).

After training, to test the reliability of the network model, some logging data and corresponding core and thin section data for Moxi21 were put into the network.

The results show the predicting accuracy for Moxi21 is 81.36% (Fig. 6b), which proves the dual neural network model has robustness for predicting both lithology and sedimentary microfacies.

Prediction

Lithology and sedimentary microfacies

After identification with all wells, several comprehensive columns were established (Figs. 7–8). The prediction results show that the sedimentary microfacies of the Member IV in the Moxi gas field are mainly laminated stromatolite boundstone microfacies, siliceous laminar boundstone microfacies, and thrombolite boundstone microfacies, with a small amount of micritic dolostone microfacies. At the same time, the prediction results show that the siliceous laminar boundstone microfacies are widely developed in the Member IV in the Moxi gas field.

Vertically, laminated stromatolite boundstone microfacies of the Member IV in the research zone intersect with the thrombolite boundstone microfacies (Fig. 8). The microfacies of siliceous laminar boundstone generally developed in the mid-upper part of the Member IV, and the micritic dolostone microfacies developed in the lower part of the Member IV (Fig. 8).

Laterally, the sedimentary microfacies of the Member IV have obvious heterogeneity, and pinch out significantly

Fig. 6 Results of training, predicting and testing. **A** The accuracy, loss of training set, and verification set changes with training epochs. **B** Partial comparison of coring results and prediction results of Moxi21 well, depth of 5140–5280 m; accuracy is 0.86

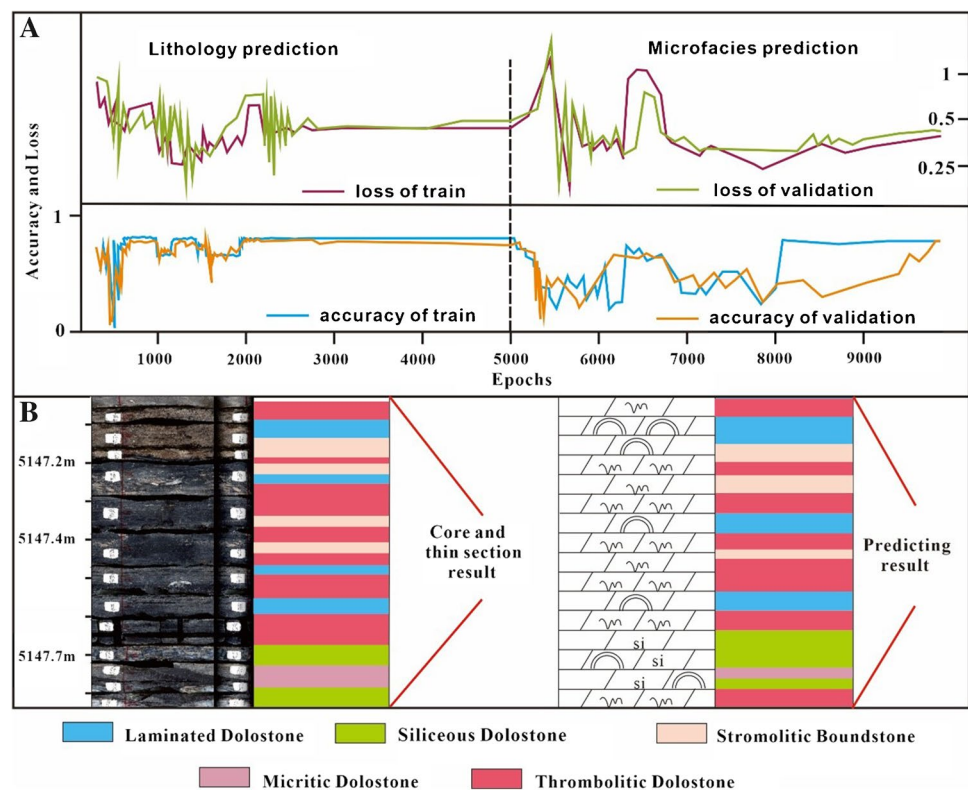
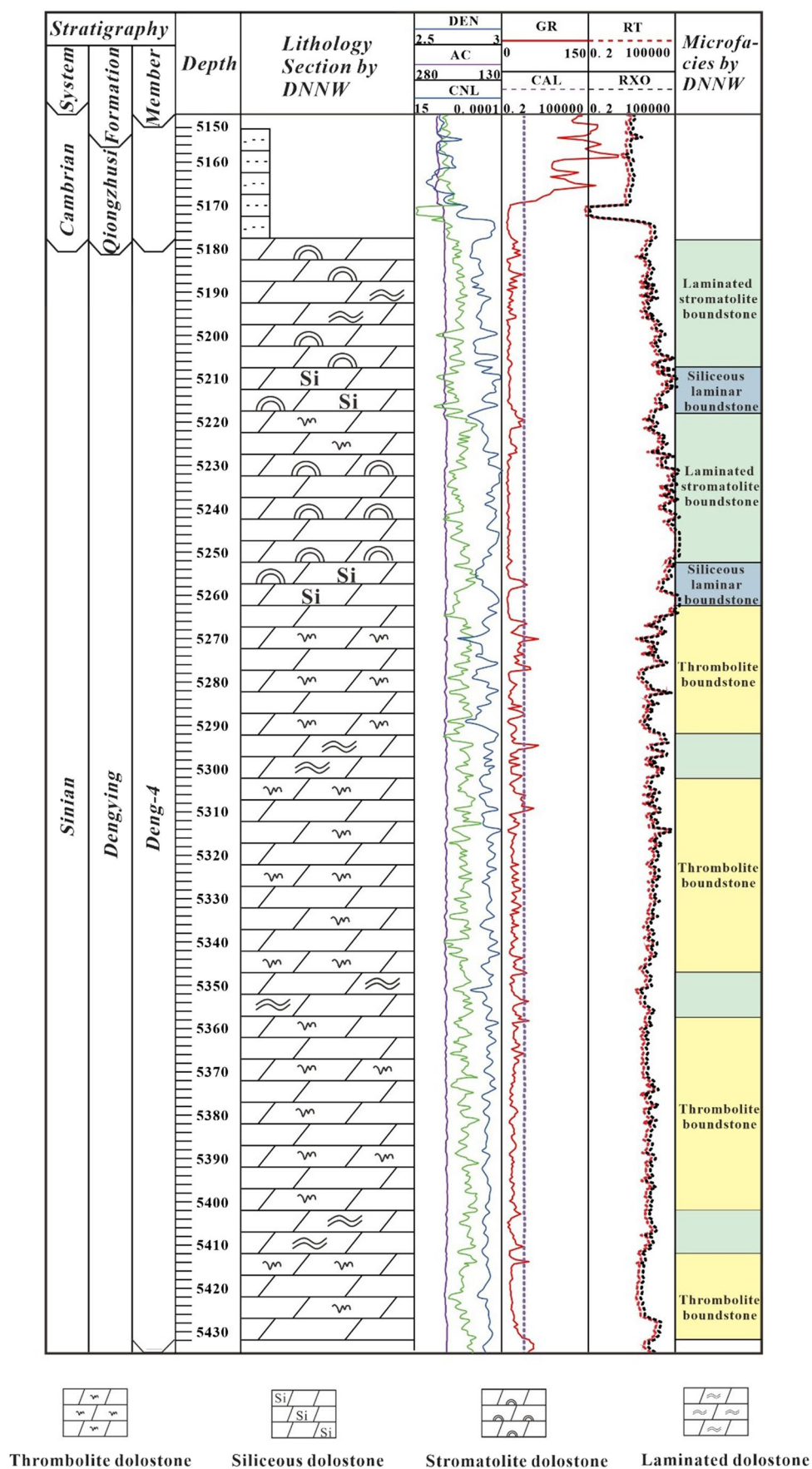


Fig. 7 Lithology and micro-facies predicting results of Moxi107, 5150–5430 m



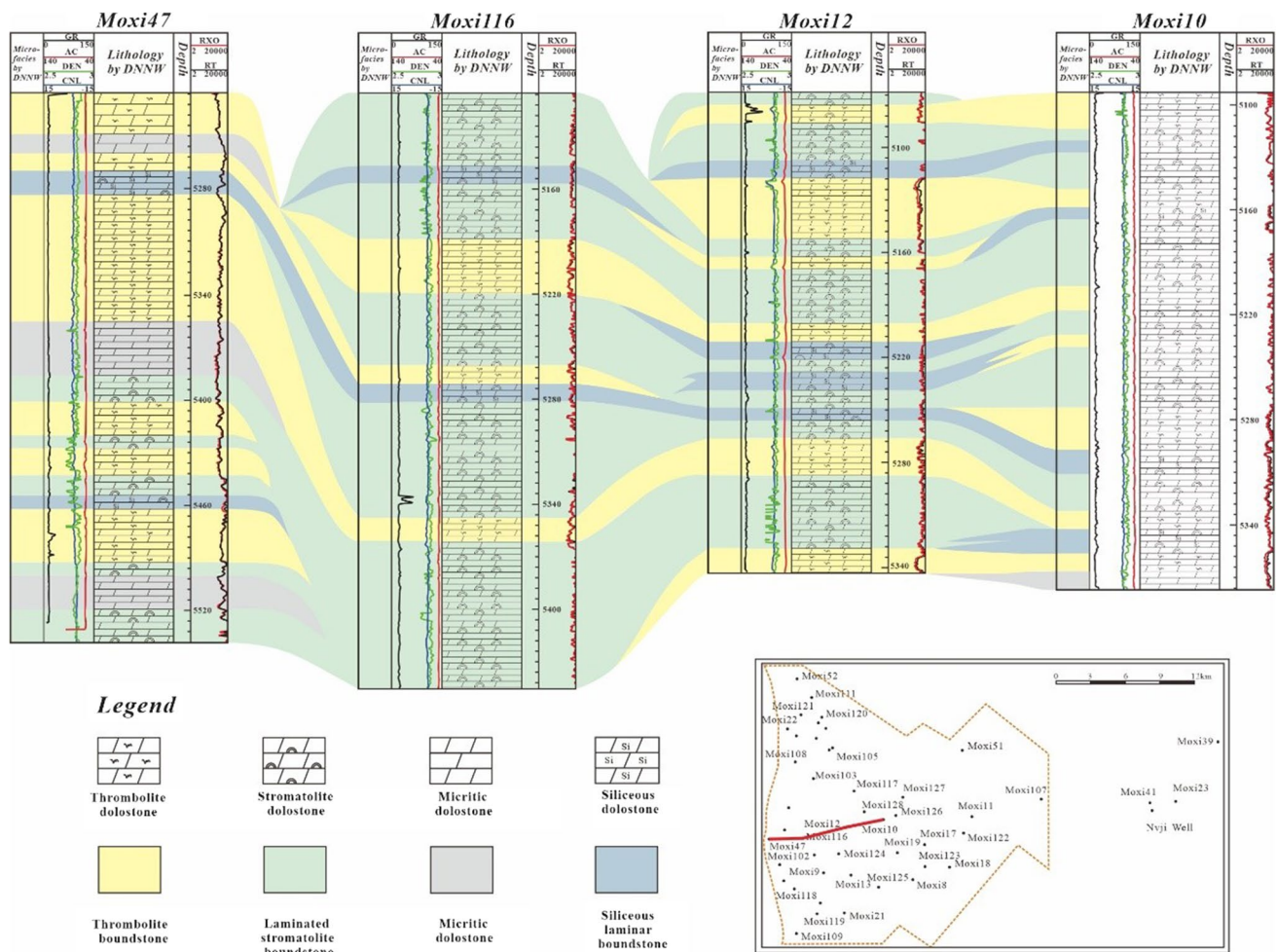


Fig. 8 Profile of connecting-well Moxi47-Moxi116-Moxi12-Moxi10

(Fig. 8). The microfacies of laminated stromatolite boundstone and thrombolite boundstone are often connected and combined in a small number of adjacent wells. The thickness of the two microfacies vary significantly in the horizontal direction, and they are combined into intermittent and undulating hills. The microfacies of the siliceous laminar boundstone and micritic dolostone are discontinuous laterally, which cannot be combined with adjacent wells and pinch out among wells (Fig. 8).

Sedimentary facies

Predicting data on microfacies in all the wells, maps of microfacies were plotted (Fig. 9a-d). The dominant microfacies are thrombolite boundstone and laminated stromatolite boundstone (Fig. 9a-d). From the top layer (1–1 layer) to the bottom layer (2–3 layer), the content of

siliceous laminar boundstone increases. Micritic dolostone facies develop sporadically (Fig. 9a-d).

After microfacies mapping, sedimentary facies mapping of the Moxi gas field were plotted from the dual neural network modeling's prediction data (Fig. 9c).

Discussion

DNNW modeling clear show the excellence in lithology and microfacies prediction. Firstly, tests with Kaggle data sets (FDNA, EALS, HRA) prove that finding mid layer and utilizing two neural network has advantage in classification tasks. With the structure of dual neural network, the training processes of DNNW models are more stable than the NNW models (Fig. 5). Furthermore, after the process of dual networks, the performances have been promoted multistagely and reach the same level with models from

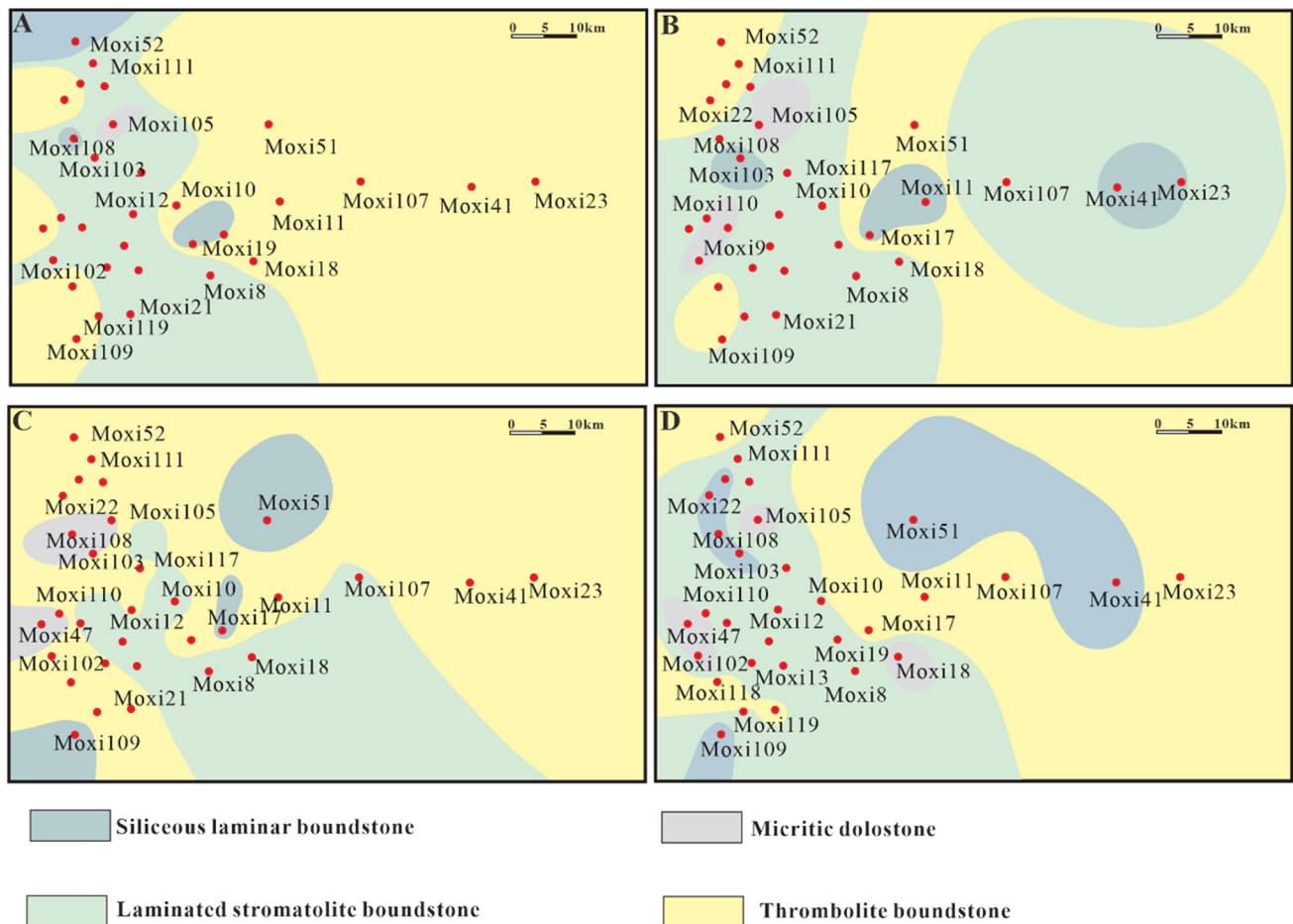


Fig. 9 Mapping of microfacies in Member IV by DNNW modeling. **A** Mapping of microfacies in Member IV, 1–1 layer. **B** Mapping of microfacies in Member IV, 1–2 layer. **C** Mapping of microfacies in Member IV, 2–3 layer. **D** Mapping of microfacies in Member IV 2–4 layer

AutoML (Fig. 5). The multistage improvements by the DNNW model outperform the traditional NNW model.

In geological research tasks, DNNW also shows superiority in well logging interpretation. Especially for oil and gas exploitation, DNNW offers a dependable petrology and microfacies reconstructing method when shortening necessary core-drilling samples. Comparing seismic data and the sedimentary facies map (Fig. 10a, b) (Jin et al. 2017), DNNW modeling results assist the sedimentary map, plotting the platform margin facies, and restricted platform facies from the predictions are coherent with the seismic data (Fig. 10a–c) (Jin et al. 2017). The two results are coherent with some different details (for instance, the platform margin plotted by the results of DNNW has similar boundary with the results of seismic data) proving the stability of the classification. Moreover, more information is accessible from the prediction of logging data. Figure 10c displays the microbial mound-shoal complex in the platform margin and microbial shoal in the restricted platform. For seismic data, the microbial mound-shoal

complex and the microbial shoal are hard to be predicted with seismic attributes, while the mound-shoal complex and shoal are essential facies for the development of excellent reservoir. Considering the high cost of seismic data collecting, the facies reconstruction with the assistance of DNNW and logging data has great potential for applying in the oil and gas exploration.

Conclusion

Through core observations and thin section identification in the Moxi gas field combined with previous research results, the sedimentary rock structure and main reservoir control factors of the Member IV in the Moxi gas field were systematically analyzed. The DNNW model compiled by Python/Tensorflow was then used to classify the sedimentary microfacies of the inner platform area and the paleoenvironment of the Moxi gas field reconstructed after testing and comparison.

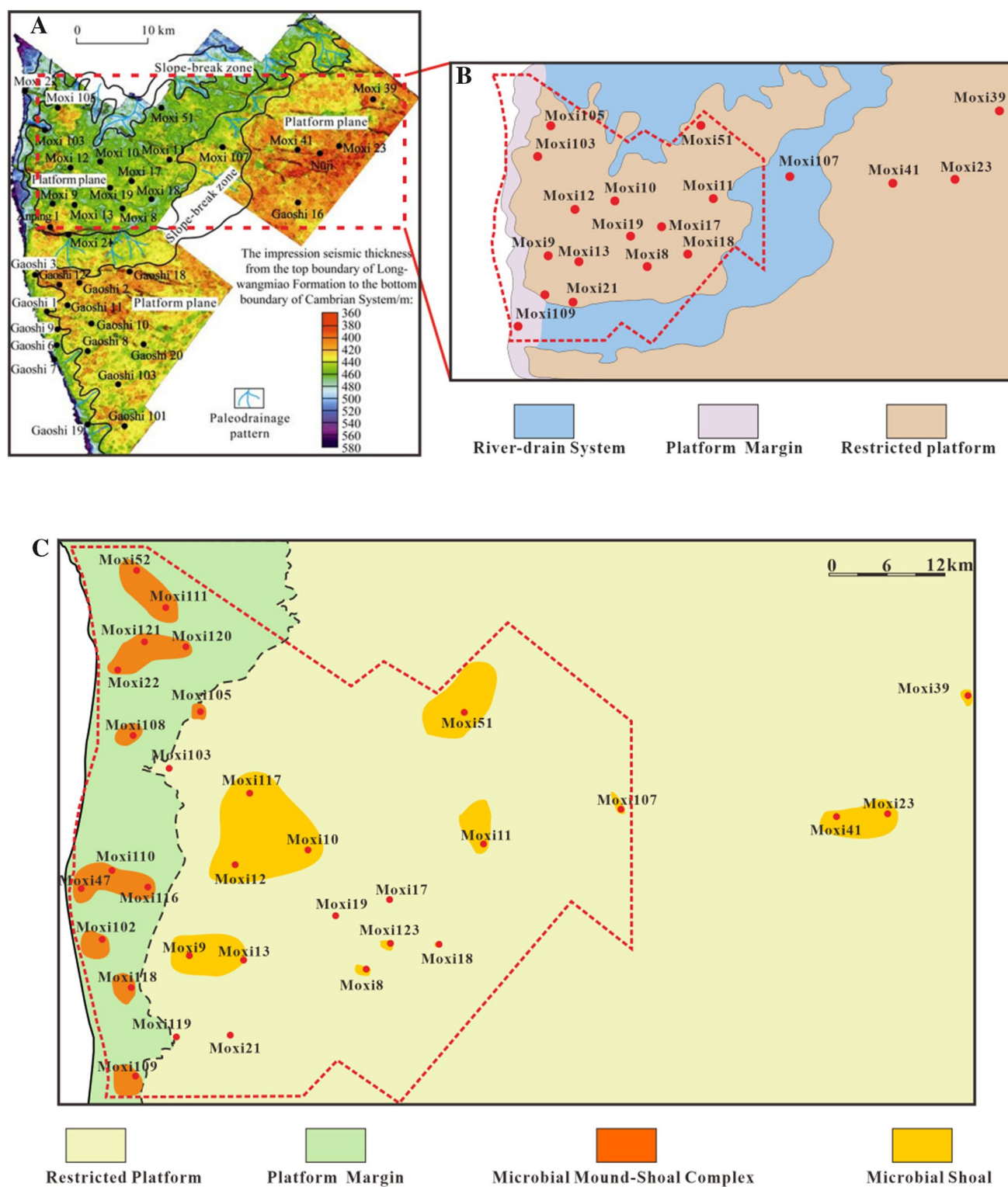


Fig. 10 Mapping of facies in Member IV, Moxi gas field. **A** Mapping of seismic thickness from the top boundary of Longwangmiao Formation (Jin et al. 2017). **B** Mapping of sedimentary facies of Mem-

ber IV (Jin et al. 2017). **C** Mapping of microfacies in Member IV, by DNNW modeling prediction

- (1) The Member IV in the Moxi gas field contains carbonate platform facies, and the sedimentary microfacies are mainly laminated stromatolite boundstone, thrombolite boundstone, siliceous laminar boundstone, and micritic dolostone.
- (2) The DNNW modeling results show laminated stromatolite boundstone microfacies, and thrombolite boundstone microfacies were developed alternately in the Z_2dn^4 in the Moxi gas field. Vertically, siliceous laminar boundstone microfacies were mainly developed in the mid-upper part of the Member IV, with only a small amount of micritic dolostone developed in the lower part of the Member IV. Horizontally, the distribution of sedimentary microfacies is obviously heterogeneous. Laminated boundstone microfacies are intermittent mounds and the siliceous laminar boundstone microfacies pinch out in the well.
- (3) Comparing with the seismic data, the sedimentary map plotted from the DNNW modeling predicting data shows uniformity, which proves the stability of the predicting results.
- (4) DNNW has strong reasonability compared with NNW. Also, AutoML shows potential for logging data processing.

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Data Availability The data that support the findings of this study are available from the corresponding author, [J.S.], upon reasonable request.

Declarations

Conflict of interest The authors declare no competing interests.

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